

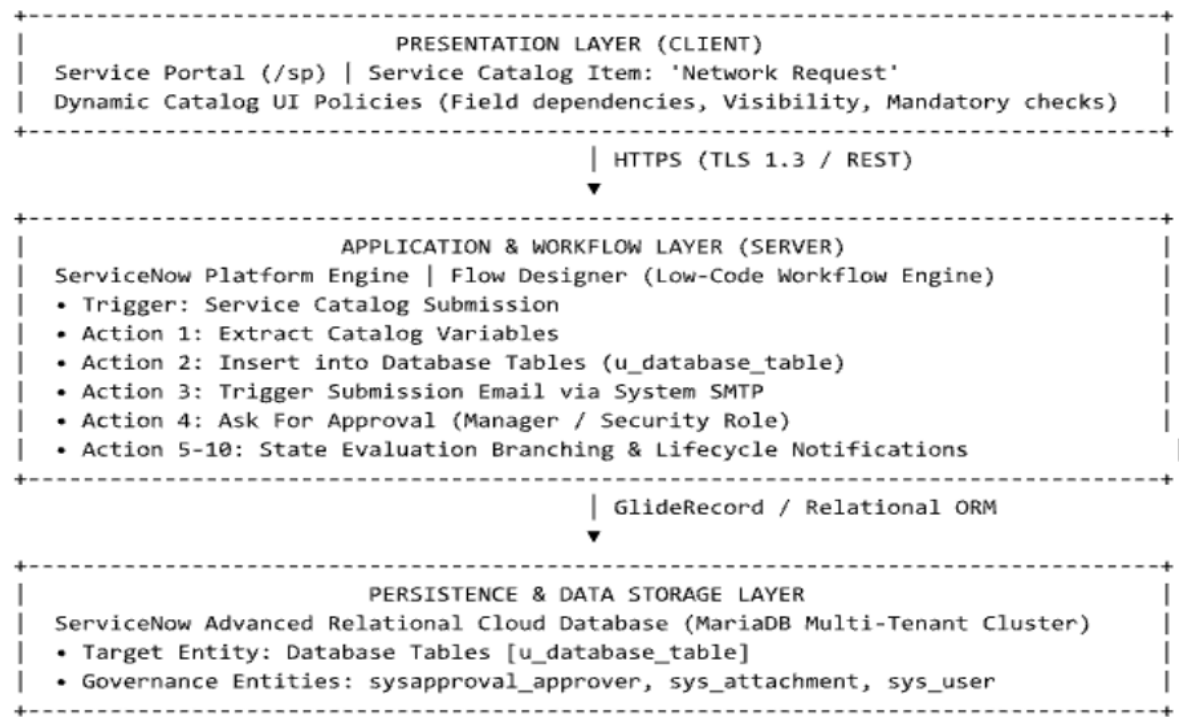
# Requirement Analysis

## Technology Stack & Technical Architecture

|              |                                                    |
|--------------|----------------------------------------------------|
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| Project Name | Automated Network Request Management in ServiceNow |

### 1. Technical Architecture Overview

The solution leverages the enterprise ServiceNow cloud platform architecture. End-user interactions occur via the modern Service Portal (/sp) executing client-side dynamic UI policies. Transactional requests trigger native Flow Designer automation, which interfaces directly with the relational database layer (Database Tables - u\_database\_table), manages approval lifecycles (sysapproval\_approver), and orchestrates automated SMTP email notifications.



**2. Table-1: Components & Technologies**

| S.No | Component                     | Description                                                                                                                          | Technology / Specification                                                          |
|------|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 1    | User Interface (UI)           | Portal interface for browsing catalog items, submitting connection parameters, and viewing request statuses[cite: 1, 7].             | ServiceNow Service Portal (/sp), AngularJS, HTML5, CSS3, Bootstrap                  |
| 2    | Client-Side Application Logic | Dynamic validation and conditional field rendering (e.g., displaying Existing ID when Type of Connection is Existing)[cite: 1, 7].   | ServiceNow Catalog UI Policies, Catalog Client Scripts (JavaScript ECMAScript 2021) |
| 3    | Server-Side Workflow Engine   | Asynchronous workflow engine that parses submitted variables, provisions database entries, and controls process routing[cite: 2, 7]. | ServiceNow Flow Designer                                                            |
| 4    | Approval Management Service   | Governance engine dispatching approval records to designated managers and capturing decisions[cite: 2, 7].                           | ServiceNow Approval Engine, sysapproval_approver relationship table                 |
| 5    | Relational Database           | Custom schema with auto-numbering to store request specifications, payment details, and approval states[cite: 2, 7].                 | Custom ServiceNow Table: Database Tables (u_database_table)[cite: 2, 7]             |

| S.No | Component                     | Description                                                                                                                  | Technology / Specification                                                     |
|------|-------------------------------|------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| 6    | Cloud Platform Database       | Underlying high-availability relational cluster supporting the ServiceNow instance.                                          | ServiceNow Advanced Cloud Relational Database (MariaDB Multi-Tenant Cluster)   |
| 7    | File & Attachment Storage     | Secure repository for uploading and referencing network diagrams, site layouts, or ID proofs[cite: 1, 7].                    | ServiceNow Attachment Engine (sys_attachment, sys_attachment_doc)              |
| 8    | Outbound Notification Service | System mailer delivering real-time lifecycle notifications (submission confirmation, approvals, and rejections)[cite: 2, 7]. | ServiceNow Email Engine (System Mailer), Outbound SMTP                         |
| 9    | Identity & Directory Service  | Directory lookups for verifying user identities, contact fields, and hierarchical manager approvals[cite: 1, 2, 7].          | ServiceNow User Administration (sys_user), LDAP/SSO Integration, RBAC          |
| 10   | Infrastructure & Hosting      | Cloud-hosted platform environment supporting catalog rendering and automated flow execution.                                 | ServiceNow Enterprise Cloud Infrastructure / Personal Developer Instance (PDI) |

**Table-2: Application Characteristics**

| S.No | Characteristics                   | Description                                                                                               | Technology / Architectural Justification                                                                                       |
|------|-----------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1    | Open-Source & Platform Frameworks | Core front-end and web frameworks used to construct responsive catalog forms and interactive widgets.     | AngularJS, Bootstrap 3.x, HTML5, JavaScript (ECMAScript 5 & modern Glide APIs)                                                 |
| 2    | Security Implementations          | Access controls, encrypted data transport, and role-based field restrictions.                             | TLS 1.3 in-transit encryption, AES-256 at-rest database encryption, ServiceNow Access Control Lists (ACLs on u_database_table) |
| 3    | Scalable Architecture             | Tiered architecture isolating user portal rendering, workflow automation, and relational data operations. | ServiceNow Multi-Instance Architecture; asynchronous Flow Designer execution queue preventing thread blocking                  |
| 4    | Availability & Resiliency         | Redundant infrastructure minimizing service downtime during catalog processing.                           | Advanced High Availability (AHA) multi-datacenter pairing, automatic failover clustering                                       |

| S.No | Characteristics          | Description                                             | Technology / Architectural Justification                                                                                                               |
|------|--------------------------|---------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5    | Performance Optimization | Low latency form interactions and fast record creation. | Platform in-memory caching, CDN edge delivery for static UI assets, indexed database keys, and client-side UI Policies avoiding excessive server calls |